# DACTIE: A Decentralized Anonymous Cyber Threat Intelligence Exchange Network

Anonymous Submission #38

***Abstract*—The cyber threat landscape is ever-evolving, requiring organizations to adopt new countermeasures, such as sharing cyber threat intelligence. Instead of reporting threats to a stakeholder through an intermediary central authority, it is more effective and anonymous for stakeholders to share information directly with each other in a decentralized network. This paper presents DACTIE, a novel concept for an efficient, anonymous, accountable, and confidential exchange of cyber threat intelligence within a permissioned, decentralized environment. To derive system requirements, we conducted three workshops with relevant stakeholders. Our implementation integrates state-of-the-art technologies, including Messaging Layer Security for traffic encryption, GossipSub for anonymous message delivery, and the first application of Short Threshold Dynamic Group Signatures for authorization. Our evaluation and comparison with related work demonstrate the feasibility and advantages of our approach. We contribute both a conceptual framework and an open-source proof-of-concept to advance secure, decentralized CTI sharing.**

***Index Terms*—P2P, Decentralized, Anonymous, Cyber Threat Intelligence, CTI, Network, Threshold Group Signature, Messaging Layer Security, MLS, GossipSub**

## 1. Introduction

With today's growing threat landscape, adopting proactive strategies, such as Cyber Threat Intelligence (CTI) sharing, seems a promising approach, recommended by cybersecurity experts [1]. Cyber Threat Intelligence involves gaining insights about threats and activities in the cyber domain by combining knowledge, context, indicators, and implications [2]. The resulting intelligence is used to proactively enhance one's strategic position against cyber threats.

Insight into the CTI of other stakeholders can be quite valuable, which is why more and more legislators are increasingly requiring companies worldwide to disclose their cybersecurity incidents within a certain time frame [3]–[9]. According to the SANS CTI survey of 2024 and 2025, these new regulations, in conjunction with audit requirements, significantly influenced 80% of companies in their CTI planning decisions [10], [11]. Therefore, sharing CTI is no longer a free choice for many companies.

Many existing CTI sharing networks tend to be centralized towards the governing entity. An example is the European Union CSIRTs network (CSIRTs network), coordinated by the European Union Agency For Cybersecurity (ENISA) [12], which is restricted to Information Sharing and Analysis Centers (CSIRTs) of EU member states only [13]. Another example are Information Sharing and Analysis Centers (ISACs), which should foster cooperation between the public and private sectors. Entities of an ISAC are often within the same industry [14], and mostly just for nationwide CTI sharing, such as a total of 23 sector-based ISACs[1] in the USA. The only international and multi-sectoral ISACs known to us are the CI-ISAC, based in Australia, and the IT-ISAC, based in the USA. While ISACs are essential, access to these networks appears to be restricted to companies that comply with strict regulations, such as signing a non-disclosure agreement. Unfortunately, limited information is available about the technologies used by ISACs to establish anonymous or private communication channels.

Research is needed to improve anonymity in CTI networks, as Abraham et al. concluded after studying the failed CTI-sharing program Automated Indicator Sharing (AIS), established by Cybersecurity and Infrastructure Security Agency (CISA) [15]. The absence of it can significantly hinder participants' motivation to share CTI and reduce its degree of detail.

In the following motivating example, we want to illustrate a use case similar to Wagner et al.'s: *"Anonymous sharing is imperative in certain circumstances when a stakeholder does not want to reveal yet that their system was breached, but wants to share the information with other stakeholders."* [16]. Figure 1 depicts the limitations of meeting regulatory requirements by sending CTI to a centralized authority (orange) and highlights a direct send path through our decentralized solution (blue).

### 1.1. Motivating Example

We assume in this network three exemplary peers representing a possible group of participants: a federal agency concerned with national cybersecurity, a company that uses software A, B, and C, and a victim company under attack. The victim discovers that the most relevant clue to the attack relates to a software exploit in "Software B".

Now it must legally report this incident to the federal agency, which first classifies it and then sends it to the companies it believes could use this information. This communication path is depicted with orange arrows. The main drawbacks of this traditional data flow are, first, that

1. https://www.nationalisacs.org/

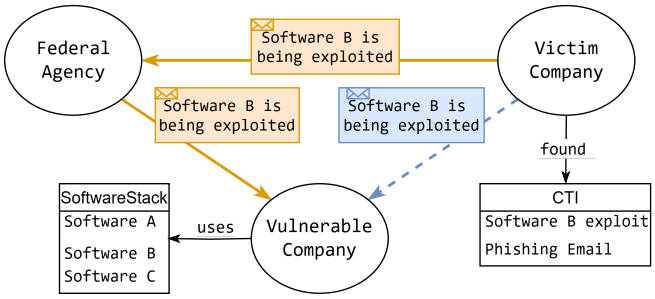



Figure 1. Network with three peers and two communication paths (Orange/Solid = current state-of-the-art, Blue/Dashed = proposed by us).

the agency is likely uncertain which peers to forward this message to because it lacks information about other peers' software stacks. And second, this approach hinders real-time information exchange because of the involvement of an intermediary.

The solution to this would be for the victim to use direct, decentralized communication to reach all affected parties, as illustrated by the blue arrow. However, the challenge remains that the victim company is unaware which peer is using the vulnerable software. This is confidential information that others will not broadcast within the network, requiring receiver anonymity. Additionally, the victim does not want to get in trouble for reporting the exploit. Possible threats include legal concerns regarding the vendor of "Software B" or potential damage to their reputation due to a cyber attack, requiring unlinkability to the origin publisher. While communicating over an intermediary, companies can never opt out of being known as the CTI source, at least by the intermediary, which is not always wanted.

This use case creates yet unresolved challenges for anonymous, decentralized CTI sharing. Therefore, we contributed to solving it.

### 1.2. Paper Structure and Contributions

We first obtain requirements for CTI sharing from real-world stakeholders and scientific work in Section 2. Please note that our contribution is not a socio-economic study. However, our workshops confirm that anonymity is a key enabler of participation and yield four high-level requirements: efficiency, anonymity, accountability, and confidentiality (R1–R4; cf. Table 2). To the best of our knowledge, no existing decentralized implementation tailored to CTI sharing simultaneously addresses all of these requirements.

Our technical contribution is the concept and implementation of our Decentralized, Anonymous, CTI Exchange (DACTIE) network designed to satisfy R1–R4. Section 3 introduces the technical background; Section 4 presents the design; Section 5 describes the implementation; and Section 6 reports our evaluation results. Section 8 discusses related work, and Section 7 discusses the results and future work before we conclude in Section 9.

## 2. DACTIE Requirements

We conducted a series of three full-day workshops over five months with stakeholders of a potential CTI sharing exchange. Participants were cybersecurity professionals and decision makers from industry, government, and critical infrastructure operators. In total, approximately 50 professionals from 25 organizations took part.

During the workshops, stakeholders were asked about their goals, perceived challenges, and requirements related to CTI sharing. The discussions were documented and later analyzed to identify recurring themes. We then compared these themes with existing scientific work to validate the identified issues and derive additional requirements for our contribution: A technical implementation of a CTI exchange network aligned with stakeholders' needs.

### 2.1. Stakeholder Goals and Challenges

Table 1 summarizes the main goals (G1–G9) and challenges (C1–C11) identified in the workshops. In brief, stakeholders seek improved, timely, standardized, and privacy-compliant sharing (G1–G5), support and incentives (G7–G8), and cost savings (G9). At the same time, they face legal and organizational issues around data privacy and compliance, fear of exposing sensitive infrastructure data, uncertainty about trustworthiness and value, limited participation, and substantial resource demands (C1–C11). These themes are consistent with prior work on CTI sharing and incident reporting [16]–[25].

Some challenges, notably costs, resource demands, and responsibility allocation (C8, C10, C11), are largely organizational and cannot be solved by technical mechanisms alone. Over time, however, the overall effort can decrease as practices mature and experienced participants support smaller organizations such as Small and medium-sized enterprises (SMEs). Other barriers are directly amenable to technical research, in particular trustworthy and privacy-preserving CTI sharing (C1, C2, C5, C7). For example, in one survey, 12% of respondents indicated that data privacy and compliance concerns were their main reason for not using CTI platforms [18]. Addressing these issues, while ensuring that misbehaving peers can be identified and excluded even in largely anonymous environments (C3, C5, C6), is therefore central to improving adoption.

### 2.2. Stakeholder Requirements

Synthesizing the goals and challenges in Table 1, stakeholders require quick and tailored information, anonymity for disseminating sensitive information, sanctions for flawed or malicious contributions, and protection against unwanted disclosure. We derived four high-level requirements (R1–R4) for a CTI exchange network and mapped them to the identified goals and challenges in Table 2.

Future work can extend these requirements with usability aspects, relevance filtering, and trust scoring. In this

TABLE 1. GOALS (G) AND CHALLENGES (C) FOR A CTI EXCHANGE NETWORK DERIVED FROM THE WORKSHOPS.

| | Goals |
|---|---|
| G1 | **Improved cybersecurity**: Strengthen stakeholders' posture. |
| G2 | **Timely sharing**: Share threat info promptly. |
| G3 | **Reduce silos**: Improve cross-organizational communication. |
| G4 | **Privacy-compliant**: Ensure legal compliance. |
| G5 | **Standardization**: Automate information flows. |
| G6 | **Best practices**: Share experiences and lessons learned. |
| G7 | **SME support**: Provide resources for SMEs. |
| G8 | **Incentives**: Motivate participation in sharing. |
| G9 | **Cost savings**: Reduce costs via attack prevention. |

| | Challenges |
|---|---|
| C1 | **Data privacy**: Lawful, privacy-preserving sharing. |
| C2 | **Confidential data**: Fear of exposing infrastructure info. |
| C3 | **Data quality**: Risk of false or malicious contributions. |
| C4 | **Relevance filtering**: Extracting relevant CTI from large volumes. |
| C5 | **Trustworthiness**: Uncertainty about data credibility. |
| C6 | **Low participation**: Many incidents remain unreported. |
| C7 | **Transparency**: Fear of reputational or competitive damage. |
| C8 | **Intangible value**: Hard-to-quantify benefits. |
| C9 | **SME support**: Insufficient resources for SMEs. |
| C10 | **Infrastructure**: Effort to operate and maintain the network. |
| C11 | **Responsibilities**: Unclear company vs. government roles. |

TABLE 2. REQUIREMENTS (R) FOR A CTI EXCHANGE NETWORK AND THEIR RELATION TO IDENTIFIED GOALS AND CHALLENGES.

| Req. | Brief description | Goals | Challenges |
|---|---|---|---|
| R1 | Efficiency: quick CTI propagation and response. | G1, G2, G9 | C4, C10 |
| R2 | Anonymity: conceal identities of network entities. | G3, G8 | C1, C2, C7 |
| R3 | Accountability: sanction misbehaving peers. | G1, G9 | C3, C5, C6 |
| R4 | Confidentiality: prevent unwanted data disclosure. | G1, G4 | C1, C2, C7 |

paper, we focus on network efficiency and technical security mechanisms that provide confidentiality and anonymity while preserving accountability (R1–R4).

## 3. Technical Background

In the preliminaries, we briefly review network architectures and security mechanisms relevant to our design.

### 3.1. Network Architectures

Baran pioneered the definition of three network types: (1) centralized networks with a star topology; (2) decentralized networks as a mixture of star and mesh components, which we focus on in this paper; and (3) distributed networks as a complete mesh network [26]. We additionally distinguish between permissionless and permissioned networks. A permissionless network is open; any peer may join. In a permissioned or closed network, participation is restricted to authorized nodes. Because CTI can contain sensitive information, we focus on a permissioned decentralized architecture.

### 3.2. Group Signatures

Group signatures [27] provide membership authentication with sender anonymity. Only group members can sign messages; verifiers can check that a signature originates from the group, but cannot identify the specific member. A designated authority (the group manager or opening authority) can later deanonymize signatures in case of misuse, combining anonymity with traceability/accountability.

**Pointcheval-Sanders Signature scheme.** The short randomizable signature scheme of Pointcheval and Sanders (Pointcheval-Sanders Signature scheme (PS16 scheme)) [28] supports efficient signatures on multiple messages, as well as randomization, aggregation, and proofs of knowledge. Signature randomization supports anonymity and prevents linkability, while aggregation and efficiency make the scheme practical for large-scale distributed systems [28], [29].

**Short Threshold Dynamic Group Signatures.** Building on PS16 scheme, Camenisch et al. propose a short threshold dynamic group signature scheme, which we will call Short Threshold Dynamic Group Signatures (Cam20 scheme) [30]. The authors modified the scheme to distribute opening and issuance among multiple parties. Shamir's secret sharing [31] is used to distribute the opening key among a set of opening authorities, such that a threshold $t$ of them must collaborate to reconstruct the key to reveal the identity of a signer and to admit new members. This reduces reliance on a single, benevolent, trusted authority and increases robustness against compromise.

### 3.3. End-to-End Encryption

Modern end-to-end encryption (E2EE) schemes such as the Signal protocol [32]–[34] are well established for pairwise messaging, but do not directly address dynamic group communication in decentralized settings. Messaging Layer Security (MLS) on the other hand provides E2EE between a set of peers [35]. It supports groups of arbitrary size with dynamic membership changes, while providing forward secrecy and post-compromise security [36]. MLS offers confidentiality, message integrity, and group membership authentication for group messaging and can be combined with various delivery mechanisms [36].

### 3.4. Anonymity

Wagner et al. categorized anonymity for CTI exchange into content anonymity, metadata anonymity, and data transfer anonymity [16]. In their work, data transfer anonymity refers to the encryption of CTI, which aligns with our understanding of E2EE described above. Additionally, unlinkability to messages is important.

**Content Anonymity.** Shared content should not contain Personally Identifiable Information (PII) or other identifiers that reveal an organization, its employees, or customers. Standardized formats such as Structured Threat Information Expression (STIX), OpenIOC, or VERIS, and the use of sharing labels such as Traffic Light Protocol (TLP), can support automated sanitization and controlled dissemination of CTI [1], [37].

**Metadata Anonymity.** Metadata anonymity, or network anonymity [29] aims to hide identifiers, such as the IP address and communication observability. Overlay networks

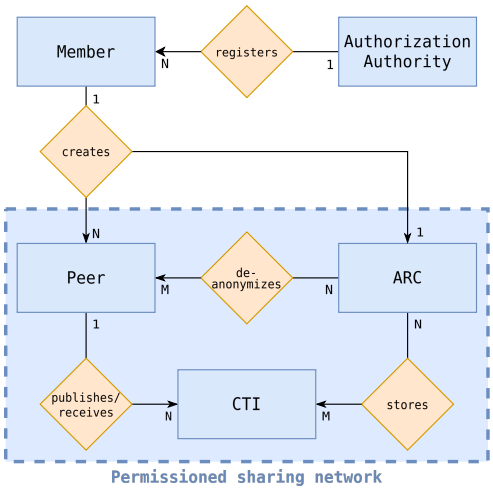



Figure 2. ERM of the peer types and their relationships.

such as Tor[2] and Mixnets [38] (e.g., Loopix [39]) route traffic through multiple relays and mix messages to obscure sender–receiver relations and timing. In our work we refer to it as Unobservability $\overline{O}$ [40].

**Unlinkability.** Chaum introduced the concept of security without identification [41], where entities use cryptographic pseudonyms that cannot be linked to real-world identities. For our work we stick to the notions of Kuhn et al. for (sender/receiver)-message unlinkability, $(S/R)M\overline{L}$ as hiding the link between a message and the sender/receiver [40].

## 4. System Design

DACTIE's concept revolves around a decentralized architecture for three reasons: First, its intermediary-free communication paths, exemplified in Section 1.1, combined with threshold signatures, enable integrity, authenticity, and non-repudiation. Second, due to the distribution of trust across multiple entities, the resilience against adversaries and censorship is strengthened [29]. And third, without a centralized point of control, the architecture allows our peers to interact without revealing their identities.

### 4.1. Peer Types

The peer types and their interaction are depicted in Figure 2. To the best of our knowledge, the state of the art for CTI sharing networks is that existing approaches are tied to a government entity. This view was also confirmed in discussions with an agency. To tailor our approach to CTI sharing, the initial registration in DACTIE involves an authorization authority.

**Authorization Authority (*AA*).** The *AA* is the only centralized component and does not exchange CTI messages. Their purpose is to vet and accept new members into the network, but they do not benefit from anonymity. As explained in our motivating example 1.1, the authority is an entity everyone trusts. This could be CISA or ENISA in a real-world scenario.

**Archives (*ARCs*.** The identity of *ARCs* is also published and any DACTIE participant can operate an *ARC*. Their role is to enable message storage and efficient querying. The *ARCs* have bigger resource costs, since they provide CTI storage capacity for the whole network. In a real-world scenario, they could be a large enterprise. This enables SME to participate without intense resources, tackling challenge *C*9 mentioned in Section 2. The archives have a special function to incentivize enterprises to offer their resources: they can partially calculate the key material required to open the signatures of misbehaving peers. When multiple archive *ARCs* up to a threshold of $t$ collaborate, they can deanonymize other peers, increasing overall trust in the network by preventing misuse of identity anonymity (*R*3).

**Member and Sharing Peers.** The publisher of CTI should be given the choice not to disclose their identity, which is one of the key requirements for our implementation. In our concept, we have separated the identity of the network members, called *MemberID*, from the actual sharing peers, the *PeerID*. The members can be seen as companies or stakeholders in a real-world scenario. Each member can operate multiple sharing peers with independent secret key pairs. The *PeerIDs* cannot be directly or computationally linked to a specific *MemberID*. As a result, message signing does not reveal the member's identity and therefore supports anonymity.

### 4.2. Threat Model and Security Goals

As a threat model, we assume that any subset of peers and up to $t-1$ *ARCs* of the network can be dishonest. The authorization authority can not be dishonest. It authorizes members' credentials and thereby prevents nation-state actors from gaining access to the network. The attackers' objectives are to eavesdrop on sensitive data, deanonymize peers, and disrupt the communication. The adversaries are assumed to be Byzantine, sending malicious or arbitrary messages, withholding messages, and prone to collusion.

**Confidentiality.** The communication is confidential if and only if the subset of peers, for whom the messages were intended, can extract useful information from a ciphertext sent as a message.

**Anonymity.** The identity of a sender *S*, e.g., the victim company from our motivating example in Section 1.1, should be unlinkable to a CTI message. An adversary can break anonymity if the Sender-Message Linking *SML*, denoted in [40], is successful more often with a probability $P_{SM\bar{L}}$, defined in Equation 4. Additionally, the topic interests of a receiver should not be derived from network observations, meaning receiver message unobservability $RM\overline{O} - |M|$. $RM\overline{O} - |M|$ ensures that no confidential infrastructure information is disclosed (consistent with *C*2). For example, if a peer subscribes to the topic *Perforce Helix Core*, it becomes apparent that this is their version control system.

2. https://www.torproject.org/

**Pseudonymity.** Pseudo-anonymity is a state where a peer's identity can be disclosed through a particular action, unlike true anonymity, where a peer's identity can not be traced. In this paper, we implemented pseudo-anonymity to achieve anonymity ($R2$) with accountability ($R3$). When a message signature is opened, the member's identity is revealed by linking all *PeerIDs* to their *MemberID* and, therefore, to all their messages. The function exists so that malicious members and all their peers can be excluded from network access (consistent with $R3$).

**Accountability.** The permissioned network remains accountable if no honest sender ever gets exposed.

**Availability.** Our implementation must guarantee the availability of messages even if multiple peers go offline, lose their connection, or if adversaries try to hold back messages of a single peer, or traffic in general.

## 4.3. CTI Sharing Groups

Another factor we designed specifically for CTI sharing is our CTI-sharing groups. As argued in Section 3.4, some messages are more sensitive and should be tailored towards a specific group of senders. Use cases can be a set of critical infrastructure companies, a private message between two peers, a subset of local peers from a different region, or a specific sector or market, protecting their sensitive CTI (e.g., classified as TLP:Amber, which has restricted disclosure policies) in front of potential competitors. Either way, it is necessary to establish a confidential channel within the network. Therefore, we divided our permissioned network into public and private MLS groups.

**Common MLS Group.** All messages sent over our network are encrypted using MLS. MLS provides group encryption which is well suited for the CTI sharing context. Instead of the $\mathcal{O}(n)$ overhead of individual transmissions, a single encrypted message can be propagated across the network in $\mathcal{O}(\log n)$ hops via tree-based forwarding. All peers in DACTIE are members of a common MLS group to broadcast public messages. The *AA* is the group owner of the common MLS group. This prevents anyone outside the network from decrypting the messages broadcasted within the permissioned network.

**Private MLS Groups.** We also allow group members of the MLS-common group to create private MLS groups with selected participants since CTI may be intended for a smaller audience. Private MLS groups are therefore nested inside the MLS-common group. The problem with the vanilla version of *openmls* is that unordered group commits, common in Peer-to-Peer (P2P) networks, can lead to group forks. Therefore, we implemented it so that only group owners can create commits, while all other members can still create proposals.

The MLS protocol, as defined in RFC 9420 [42], requires a delivery service for data distribution and peer discovery, and an authentication service with signatures for integrity protection.

## 4.4. Delivery Service

As a decentralized PubSub delivery service propagating over all peers of the network, we utilize GossipSub[3]. It is an unstructured, decentralized Publish/Subscribe (PubSub) overlay built on top of the libp2p framework. Peers exchange messages over full-message peering links. Additionally, metadata-only "gossip" is used to disseminate message identifiers to randomly selected neighbors, improving robustness and propagation efficiency [43]. GossipSub is resilient against common attacks on P2P networks and operates at a lower level compared to protocols such as Matrix[4] or P2Panda[5], avoiding unnecessary features. Compared to TAXII's PubSub mechanism, which is yet to be specified[6], it is mature and widely used in practice, e.g., in Ethereum.

In GossipSub, peers subscribe to channels of interest and track their neighbors' subscriptions. We modified this subscription mechanism to allow receivers to hide their interests, thereby enabling stronger metadata-anonymity guarantees for them. Multiple topics are sent over a single subscription, preventing other peers from deducing which topics another peer is interested in based on their subscription. This technique is called partitioned channels. The idea of partitioned channels is adopted from the messenger protocol Status[7].

Gossip protocols are designed to provide source anonymity [44]. The fundamental idea behind *gossiping* is to relay messages through multiple propagation peers. For the receiver, it isn't easy to distinguish whether a message originated from the propagator or the original publisher. To apply this concept to GossipSub, the protocol must be correctly configured. In the original GossipSub library, messages are flood-published [8], meaning that a peer sends a message to all known other peers. If an attacker peer is in the Distributed Hash Table (DHT) of a publisher, $S$, the attacker will always get the message directly from $S$, removing anonymity. Consequently, we disabled floodpub in our implementation..

**Bootstrapping.** Because GossipSub is unstructured, it requires a structured peer-discovery mechanism as an address store [45]. DACTIE utilizes libp2p's Kademlia: A structured and decentralized P2P overlay network, which uses a DHT. Each node maintains a routing table of *PeerIDs* mapped to their addresses [46]. New peers seed their Kademlia routing table with a known "bootstrap" peer, which is the authority's *PeerID*. They then rely on the identify behavior to progressively add the listen addresses of discovered peers to the Kademlia DHT. The peer identifier (PeerID) of Kademlia consists of a random 160-bit string chosen by the node [47].

3. https://github.com/libp2p/.../gossipsub/README.md
4. https://github.com/matrix-org
5. https://p2panda.org/
6. Chapter 6 of TAXII 2.1 Specs
7. https://specs.status.im/spec/10
8. https://docs.rs/.../struct.GossipsubConfig.html

### 4.5. Authentication Service

In total, we employ three different types of signatures within the network. The first two are member and peer signatures, which are Ed25519 elliptic curve signatures, and of which the *PeerID* and *MemberID* are derived. By having separate signatures, peers are not directly linked to a member. The third type of signatures is the Threshold Group Signatures as described by Camenisch et al. [30], which uses the PS16 scheme as a foundation (Section 3.2). It produces relatively small signatures, making it efficient, and is used for member authorization and for tracing messages back to an identity when a threshold of archives opens them.

Chaum and van Heyst proposed the group signature scheme, the only signature family that enables anonymous yet traceable signatures. As property, they defined that a group signature can be "opened" to reveal the signer's identity [27]. In our concept, a threshold of all archive peers can "open" a peer's signature, rather than a single authority. This reduces the risk of abuse because multiple entities would have to collude. Opening a signature is an edge case to protect the network from malicious peers. This weakening of anonymity is a trade-off for accountability, fostering trust within the network by allowing malicious messages to be traced back to their origin. Therefore, publishers are held accountable, which is our requirement *R*3.

Each message sent over the network must be signed with a valid Cam20 scheme group signature to be propagated and accepted by other sharing peers. An *AA* validates members' identities and registers them in the network. Bootstrapping leakage of new *PeerIDs* is possible, but irrelevant as the *MemberID* is not visible in the CTI exchange network.

### 4.6. Message Storage and Blockchains

Without message storage, peers that are temporarily offline may otherwise miss important updates. Centralized systems can conveniently buffer messages at a central server. In contrast, P2P solutions such as BitTorrent or Interplanetary Filesystem (IPFS)[9] distribute data across all network members, which makes efficient querying more challenging.

Blockchains are not designed to store large datasets containing CTI and often rely on off-chain storage solutions. This design results in significant transaction overhead, which is undesirable for efficient sharing. While other papers use IPFS as a storage layer solution [48], [49], we have chosen dedicated archive peers. *ARCs* directly receive all messages exchanged via GossipSub, eliminating the need for an intermediary between IPFS and our peers. Furthermore, archive peers leverage efficient database technologies, enabling specific queries on the accumulated data and better use of machine learning applications than distributed storage systems such as IPFS. The ability for any participant to initiate an archive also mitigates the risk of a single point of failure.

9. https://ipfs.tech/

Another inefficiency of blockchain approaches arises from the use of a consensus algorithm. Achieving consensus on all shared CTI may not be necessary, as not all peers require the same information, and the order of message arrival is irrelevant as long as the process is swift and reliable. Thus, consensus introduces unnecessary protocol and storage overhead. Utilizing the underlying Gossip Algorithm directly for transaction sharing across the network is a more effective approach. This rationale also applies to IPFS. Additionally, the costs of public blockchains, such as Ethereum, are often underestimated. Consequently, we sought a non-blockchain, non-IPFS approach for our research, leveraging a Gossip Algorithm.

In DACTIE, each *ARC* hosts its own database, introducing redundancy because multiple *ARCs* can store the same data. This redundancy ensures that sharing peers do not have to rely on the trustworthiness of a single archive. Additionally, not all archives will have the same messages at the same time. But this delay is tolerable and redundancy beneficial in our view. Messages of the Common MLS Group are stored in plaintext, enabling message querying, while private MLS Group messages are stored in ciphertext, since the *ARCs* cannot decrypt them. Public CTI feeds are intended for the entire Common MLS Group as an audience.

## 5. Implementation

The DACTIE implementation is written in 6,705 lines of pure Rust. Rust is more secure than other programming languages [50] and has well-maintained libraries for MLS[10] and GossipSub[11], which we used. Moreover, we implemented the first publicly available proof-of-concept (PoC) for the Cam20 scheme adjusted to our use case.

### 5.1. Register

When members want to join the permissioned network as shown in Figure 2, they must register with the Authorization Authority *AA*. To do so, the peer creates an Ed25519 keypair and derives their *PeerID* according to the libp2p spec[12]. The *PeerID* and verification materials, such as the company's name, contact details, and other proof of identity, are sent to the *AA*. The *AA* verifies manually that the peer belongs to the respective company and is eligible to join the network. After successful verification, the GroupKey, parts of the MemberKey, such as the Zero-Knowledge Proof (ZKP) parameters, and the public keys of the *ARC* are sent to the peer. The peer then chooses a secret key *sk*, calculates its blinded version, and generates the Shamir shares with it. Each Shamir share is encrypted with an ElGamal public key of one of the *ARCs*, which the *ARCs* have generated and sent to the *AA* before. Since the shares are encrypted and *sk* is blinded, Non-Interactive Zero-Knowledge Proofs (NIZKs) are generated for each to prove that *sk* can be derived from

10. https://docs.rs/openmls/latest/openmls/
11. https://docs.rs/.../libp2p_gossipsub/
12. https://github.com/libp2p/specs/blob/master/peer-ids/peer-ids.md

all of them. Details can be found in the GJoin.U joining protocol of the Cam20 scheme [51].

In the next step, the *AA* receives all of that, plus share commits and a publicKey for the MLS group. Then it verifies the NIZKs for the blinded *sk* and the Shamir shares to confirm that they were calculated correctly. After that, the *AA* forwards each *ARC* their encrypted Shamir share, along with the NIZK and share commits. The difference between our implementation and the paper is that we have one issuer, the *AA*, and therefore only one entity where this is stored, rather than on a public ledger. In the next step, each *ARCs* stores their encrypted shares and commits after verifying them with a NIZK. In parallel, the *AA* calculates another part of the MemberKey by signing the blinded *sk* themselves, which is comparable to signing *sk* directly. Then this signed, blinded *sk* is returned to the peer, which now has all the MemberKey parts and can create the signature for signing messages broadcasted on the network. As a last step, the *AA* adds the peer to the commonMLS group using the peer's earlier sent public key. A sequence diagram is depicted in Figure 7.

Archives register differently. In our PoC, they are registered once at the beginning of the network initialization phase. To do so, the *AA* verifies the identity of the *ARC* and responds to them. After the *ARCs* are registered, the MLS common group is created, and the *ARCs* are added.

## 5.2. Create

For a peer to create multiple sharing peers, it must register another *PeerID* at the *AA*. The data flow is almost identical to a new peer joining the network: The new key material for the new *PeerID* is sent to the *AA*, which requests the public keys of the *ARC*. After receiving the *ARC* pubkeys from the *AA*, the peer uses their existing group signature *MemberKey* to create new shares for the new *PeerID*. The *AA* forwards them to the *ARCs*, which store them. A peer with two *PeerIDs* has three shares at each *ARC*: two for the *PeerIDs* and one for their *MemberID*. Additionally, the *AA* adds the new *PeerID* to the MLS common group.

## 5.3. Publish/Receive CTI

To publish and receive CTI, our network covers four different ways to send messages as illustrated in Figure 3. The default setting is public/anonymous. To broadcast a message publicly, a peer subscribes to a GossipSub channel. This channel is signed with the payload of the Cam20 scheme signature GroupKey. With the signature, it is verified that the origin sender *S* is an authorized part of the network. If the signature is invalid, another peer will not propagate the message. This error can be reported to the *AA* or *ARCs* to expose malicious behavior.

The signature will always differ, even when the same peer broadcasts the same message over the same channel, due to the scheme's randomization. See Figure 8 for a console output portraying this behavior. Through GossipSub's partitioned channels, topics are addressed and forwarded by the last sender. Therefore, it remains unclear whether the *PeerID* that propagated the message to the receiver is the message's origin. The payload is end-to-end encrypted with the MLS common group key, enabling protection against external adversaries.

To create attribution towards a peer's *MemberID*, it can be added to the message payload. This may be necessary to foster trust in specific CTI exchanges that require attribution. Editing the message payload of another peer and pretending to be a particular member renders the signature invalid, leading to a potentially reported message towards the *AA* or *ARCs*. Both the attributed and anonymous scenarios are possible inside a private MLS group, which any peer can freely create. The message header, second to the *PeerID*, is addressed to the private MLS GroupID to which the peer wants to send the message, instead of a public channel. The encryption key of the payload changes, and only members of the targeted MLS group can decrypt it.

## 5.4. Deanonymize

Any peer can request the archives to open a signature to expose a network member. This process starts with sending a signature that a peer wants to open to the *ARCs*. In a more realistic scenario, there would be additional evidence of misbehavior. The *ARC* can then decide if the proofs are sufficient and if they want to deliver their parings to the peer. Pairings are computed by iterating over all Shamir shares and commits, using each share's signature to derive the pairing. The parings are combined with the *PeerID* or *MemberID* and sent to the peer. This is highly inefficient, since the signature must be tested on each stored share. However, this inefficiency is another wanted factor that makes opening a signature an edge case.
By combining the pairings from multiple *ARCs*, the peer or *AA* can verify whether the signature matches the *sk* contained in the pairings without disclosing *sk*. Nevertheless, all *PeerIDs* belonging to a *MemberID* can be exposed, and therefore all their messages can be assigned to the identity behind it.

## 5.5. Store

To store the CTI, we use PostgreSQL 16.3 databases on a Debian 12.2.0 Server. The archives subscribe to all GossipSub channels, ensuring they receive all messages sent over the network. The received CTI is stored in the database and can be accessed by every peer. *ARCs* are required to just accept queries from peers with a valid signature. In the case of public CTI, archive peers play a crucial role in decrypting and extracting the content, enabling other peers to efficiently query all public CTI. Private messages, on the other hand, are only intended for private group members and cannot be decrypted. Storing them in an archive peer primarily serves as backup in case of disconnection. Archive peers only need to store private messages for a limited time.

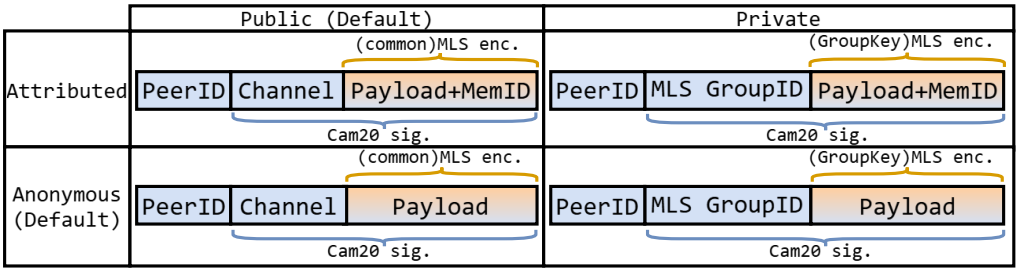



Figure 3. CTI sharing exchange message formats (private/public/anonymous/attributed).

## 6. Evaluation

This section discusses how we fulfill our requirements R1–R4.

### 6.1. Anonymity

Our anonymity security goal, defined in Section 4.2, requires sender-message unlinkability $SM\bar{L}$ and receiver message unobservability $RM\bar{O} - |M|$.

**Receiver Message Unobservability.** Our network provides metadata anonymity for receivers, or $RM\bar{O} - |M|$, through partitioned channels. Using a hash function to select the channel for a topic makes it difficult for attackers to trace specific messages to a single peer. Peers can subscribe to channels even if the topic they are interested in is not actively exchanged, while an attacker listens. Topics that a receiver is not interested in can be automatically filtered out. Consequently, even if only two peers join a channel, an attacker will not know that they are communicating. However, receiver anonymity is influenced by the number of topics per channel. The maximum number of available channels, $C_{max}$, determines how many participants can be on each channel, thus creating a trade-off between efficiency and anonymity. This threshold can be adjusted during the network's runtime as peers join or leave. Topics and their amount are randomly distributed across all channels using a hash function, as described in the following equation:

$$C = \mathscr{H}(topic_descriptor) \mod C_{max} \tag{1}$$

**Sender-Message Unlinkability - Theoretical Bounds.** $D$ denotes the mesh degree, i.e., the number of peers with which each peer maintains full-mesh connections. Let $N_c \subseteq N$ denote the subset of $N$ network nodes of size $|N_c|$, subscribed to a channel $C$, and let $S \in N_c$ denote the message source - not a peer forwarding $S$'s message. We define the following mutually exclusive linkability events:

- $\mathscr{E}_{\text{dir}}$: the attacker receives the message *directly* from $S$ via a full-mesh connection,
- $\mathscr{E}_{\text{ind}}$: the message is *not* received directly, yet can still be traced back to $S$ through relay hops.

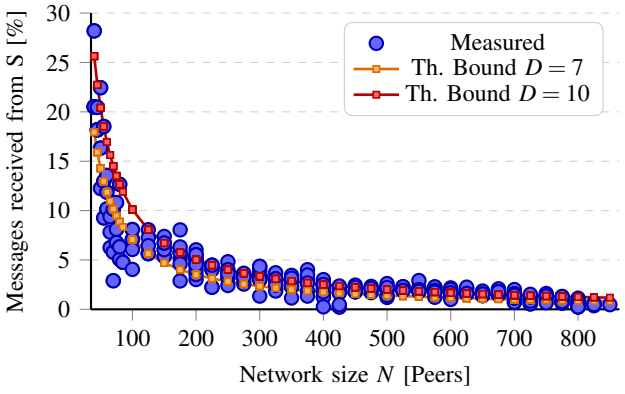



Figure 4. Anonymity experiment results with theoretical bounds for fanout $D \in \{7, 10\}$

An attacker's probability of receiving the message directly from $S$ is the full-messaging connections $D$ relative to the remaining $|N_c| - 1$ subscribers:

$$P(\mathscr{E}_{\text{dir}}) = \frac{D}{|N_c| - 1}. \tag{2}$$

If the message was *not* received directly from $S$, it may still be linked to $S$ through indirect propagation over $H$ hops with branching factor $B$, where $B$ denotes the number of peers to which a node forwards a novel message upon first receipt. Since each flooding step multiplies the number of reachable paths by $B$, the conditional probability of indirect linkability is:

$$P\left(\mathscr{E}_{\text{ind}} \mid \overline{\mathscr{E}_{\text{dir}}}\right) = \frac{1}{B^H}, \quad B \approx D. \tag{3}$$

Applying the law of total probability over the two mutually exclusive events in (2) and (3), the overall sender–message linkability probability is:

$$P_{\text{SM}\bar{\text{L}}} = \frac{D}{|N_c| - 1} + \left(1 - \frac{D}{|N_c| - 1}\right) \cdot \frac{1}{B^H}. \tag{4}$$

**Experimental Bounds.** In DACTIE, we implemented the anonymity experiment shown in Figure 4. We inserted our experimental values into equation 4; using the values 6 to 10 for $D$; $B = D - 0.5$; and $H$ was 34.67% of $|N_c|$ on average. For $D = 7$, the theoretical bounds were surpassed 25.48%

| Parameters | Values HF | Values DACTIE |
|---|---|---|
| Nb. of Channels | 1 | 1 |
| Complexity | 30 bytes * TAR | 760 bytes * X |
| Resources | 32 vCPUs | 4 vCPUs |

TABLE 3. HYPERLEDGER FABRIC COMPARED TO DACTIE

of the time, while with $D = 10$, the theoretical bounds were surpassed 62.42% by the measured experiments. This demonstrates that $D$ impacts highly whether our experimental values surpass our theoretical bounds for $P_{SM\bar{L}}$.

Even if an attacker guessed $S$ correctly with a probability of $P_{SM\bar{L}}$, a single message is only linked to the *PeerID*, which is a pseudonym and does not contain any identity-revealing information, therefore guaranteeing $SM\bar{L}-P$, meaning linkage to a pseudonym. Members can create new *PeerIDs* for every new message to increase unlinkability.

**Scalability.** Our anonymity experiment shown in Figure 4 also yielded insigts on scalability. For the experiment, we spawned up to 800 containers (peers) on a server with the following specs: 32 vCPUs (AMD EPYC 7763), 15 GiB RAM, 1 TB HDD, and Ubuntu Linux 5.15.0-168-generic (x86_64).

A concrete example is spawning networks with 700 to 800 peers in 25-peer steps, with 4 runs each, resulting in a total of 20 runs (4*700, 4*725, 4*750, 4*775, 4*800). After each run, the script idled for 4 minutes to allow the peers to exchange messages, then sent a measurement message afterward. All runs were executed in 1h43m, of which 1h20m (20runs * 4min) were idle time. With a lower idle time, some attempts failed. However, 4 minutes of idle time achieved a 100% success rate across 300+ runs, with no failed builds. This demonstrates that huge DACTIE networks can be reliably built and run in roughly five minutes.

## 6.2. Efficiency and Performance

**Macrobenchmark.** The most impactful aspect of efficiency, meaning how quickly a message can be sent from a propagation source to its recipients, is latency. The only scientific work we know of that evaluates anonymous CTI sharing in a permissioned decentralized network is [52]. Their simulation is not yet developed, but they estimate a throughput of 40 transactions per second, assuming that each of 3,338 critical infrastructure companies sends 10 transactions per day. They refer to the Hyperledger Fabric (HF) benchmark [53], also used by other related work [54], [55], which we will evaluate against as a baseline.

To compare our evaluation, we also use a single channel, but fewer vCPUs, and our message length is higher, as shown in Table 3. Since we do not have transactions, we compared message complexity per second rather than the Transaction Arrival Rate (TAR). From Figure 4 of [53], we derive that, e.g., with a TAR of 25 transactions per second and a block size of 30 bytes, we achieve a similar complexity of 750 bytes per second, compared to the size of one DACTIE message of 760 bytes per second. (Another example would be $50TAR * 30bytes = 1500bps$ compared

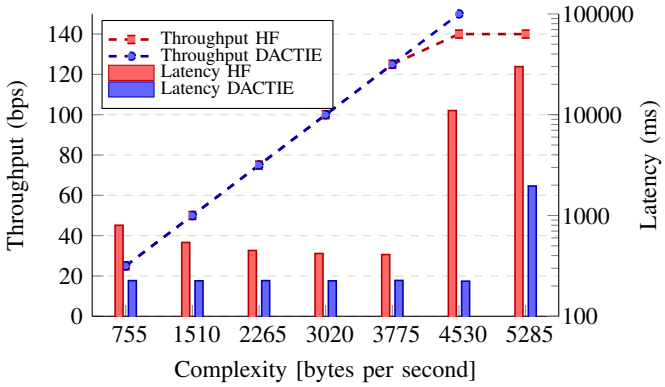



Figure 5. Impact of block size and transaction arrival rate on performance (Block size: 500)

to $760bytes * 2 = 1520bps$, ...) Therefore, we can compare DACTIE's latency to the HF's latency using their "Latency block size:30" metric of their Figure 4 in our Figure 5. We also conducted the experiment in two scenarios: one where the sending and receiving peers were on the same server, and the other where the sender and receiver were on distinct servers, and achieved similar latency.

Our latency evaluation shows that we achieve approximately half the latency of HF, which most related work is built upon [52], [54], [55], demonstrating that DACTIE's GossipSub network is effective. Another research paper, which also uses GossipSub as an underlying communication protocol, conducted an experiment on the Ethereum blockchain and showed that, in 98% of cases, a message can be sent to all recipients in less than 4 seconds [56].

**Microbenchmark.** By being the first CTI sharing concept to leverage the group encryption properties of the MLS protocol, our approach has a clear efficiency advantage over comparable solutions that use legacy group encryption, where each message must be encrypted separately for each recipient. The efficiency of encryption with MLS has been tested in former research [57]. A paper with a design even more similar to ours by Soler et al. used the same open-source libraries as we did to experiment with MLS operations provided by GossipSub. Although GossipSub exhibits twice the latency of a centralized MQTT service, it stabilizes at around 2 seconds in networks ranging from 512 to approximately 1,000 users due to the higher bandwidth of larger groups. The authors also measured the CPU time required to generate and process MLS messages and concluded that this time is negligible compared to latency [36].

After the payload encryption with MLS, messages sent over DACTIE are signed using the Cam20 scheme group signature [30]. We expect a negligible performance impact, since signing and verifying messages requires only a small number of operations, and leave this for future work. In contrast, de-anonymizing the identity of the message's origin is inefficient, since each archive must compute for each Shamir share. However, opening a signature is rare; therefore, this inefficiency is justifiable.

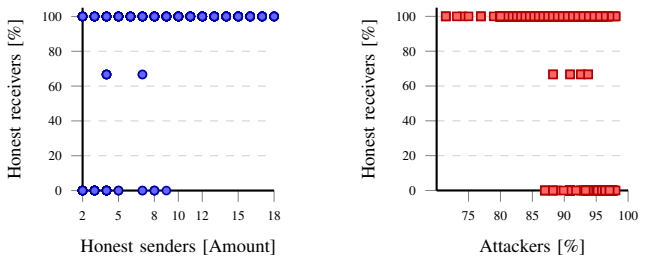



Figure 6. Eclipse experiment results

### 6.3. Accountability and Collusion Discussion

The threat model for this scenario requires a threshold $t$ of all archive peers *ARC* to collude and disclose the identity of a target participant. Rather than recommending a static value for $t$, we propose a network-adaptive threshold $t$ to achieve a more efficient consensus [58]. In our used scheme [51], no lower and upper bounds for $t$ were given. The lower bound considers corruption, fault, and collusion resistance [59], [60], while the upper bound considers availability.

**Lower Bound.** The lower bound of $t$ can be derived from the observed composition of the set of *ARCs*. Let $n_{\text{ARC}} = |ARC|$ denote the total number of *ARCs*, and let $G = \{G_1, G_2, \ldots, G_k\}$ be a partition of $n_{\text{ARC}}$ into $k$ coalitions of peers sharing a common interest (e.g. market membership), which we expect to be more prone to collusion. Define the *dominant coalition size* as:

$$g = \max_{i \in [k]} |G_i| \tag{5}$$

The lower bound needs to be at least $t \leq g + 1$ to permit the dominant coalition $G_i$ — with $|G_i| = g$ — to unilaterally satisfy the opener quorum.

**Upper Bound.** The practical upper bound imposed by the total number of openers $n_{\text{ARC}}$, the admissible range for the threshold is:

$$g + 1 \leq t \leq n_{\text{ARC}} - 1 \tag{6}$$

Raising $t$ to $n_{\text{ARC}}$ -1 reduces availability, as more openers must actively coordinate to produce a valid opening. The choice of $t$ within this range therefore reflects a security–availability trade-off governed by the use case's trust assumptions.

In a permissionless network, Sybil attacks in which *ARCs* exceed $t$ of $n_{\text{ARC}}$ are possible because a subset of malicious *ARCs* can be created. In our network, creating a new *ARC* peer is not trivial. First, the identities of *ARC* peers are made public to create transparency. Second, there is also a significant technological hurdle, since every peer's Shamir share must be encrypted for each *ARC*. Because of the latter, all key material for all peers must be renewed if a new *ARC* joins DACTIE. Sybil attacks are therefore prevented by the authority's membership verification process.

### 6.4. Security

We divided our security goals in Section 4.2 into confidentiality, anonymity, accountability, and availability. Anonymity and accountability were already discussed in detail in the former subsections, leaving confidentiality and availability for this section.

**Confidentiality.** The overall confidentiality of our scheme depends on the security of MLS. The MLS protocol provides guarantees for "confidentiality, integrity, and authentication for the application data" [42], meaning all data exchanged between group members. Additionally, the MLS protocol, as well as the group signatures, remain unchanged, ensuring the security proofs of their respective papers are valid [30], [61], [62].

**Availability.** CTI availability for receivers is guaranteed due to our redundant storage design across *ARCs*. Additionally, former research has assessed the security of GossipSub by examining the effectiveness of countermeasures such as peer scoring, random peer selection, and flood publishing to strengthen it against common P2P attacks, including Sybil and eclipse attacks [63].

We modified the open-source implementation by disabling *floodPub* in our network to strengthen DACTIE's source anonymity guarantees. Flood publishing serves primarily as a defense against eclipse attacks. Therefore, we tested the feasibility of eclipse attacks in our network. Let $H$ denote the set of honest peers and $A$ the set of attacker peers. The total network is defined as:

$$N = H \cup A, \quad H \cap A = \emptyset \tag{7}$$

An honest victim peer $V \in H$ is selected, from which $A$ does not propagate messages. If $V$ only has full-message peering connections with $A$, this simulates an eclipse attack since no published message will reach any peer in $H$. We chose different values $n \in \{12, \ldots, 125\} \subset N$. At least two peers needed to be in $H$ to measure if a message was received by another peer in $H$, who is not the sender $V$.

Our results yielded two important conclusions depicted in Figure 6. In the first graph, we compared the number of message receivers in $H$ to $H$'s total size. In over 800 runs with at least 10 peers in $H$, no eclipse attack was successful. Let $\alpha = \frac{A}{n}$ denote the fraction of attacker peers in the network. We concluded that, regardless of $n$, all eclipse attacks are unsuccessful when $\alpha < 0.86$. When $\alpha > 0.86$ and at least 10 peers in $H$, the network resists eclipse attacks as well. We conclude that our network is sufficiently resilient to eclipse attacks, guaranteeing availability, even with floodpub deactivated.

## 7. Discussion

This work takes some socio-economic aspects gathered from the workshops into account. However, continuous active participation from the network members and their contributions are required. An open research topic remains how to incentivize potential competitors, for example, in

industrial manufacturing, to collaborate and share their CTI with a broader audience.

Further future work for our implementation encompasses metadata anonymity for senders, $SM\overline{O} - |M|$. Currently, $SM\overline{L}$ and $RM\overline{O} - |M|$ are ensured, as long as accountability is not compromised, which is sufficient for most use cases. Nevertheless, we are confident that our implementation is compatible with existing state-of-the-art technologies, such as TOR or the Nym Mixnet, to ensure $SM\overline{O} - |M|$.

However, we are aware that our proposed solution does not match all existing use cases for anonymous information sharing. We determined in our requirements that anonymity and accountability are essential for the network's acceptance by our workshop participants, the stakeholders. Therefore, our concept is only feasible for stakeholders with similar requirements.

Another implementation idea is to adjust the network to support multiple authorities. For our use case, it was sufficient to demonstrate the interplay of our technologies for the realization of DACTIE with one authority. As explained in the motivating example in Section 1.1, we focused on improving the current state of the art, which tends to be centralized around the governing entity. To the best of our knowledge, no open-source, decentralized solution exists to share CTI in a decentralized manner.

Our approach can also be used to share sensitive data beyond CTI, as it emphasizes establishing technically secure communication between peers. Peers can share CTI or other private data publicly within the permissioned network through partitioned channels or privately within groups, while meeting the requirements of confidentiality, anonymity, accountability, and availability.

## 8. Related Work

DACTIE is the first efficient, anonymous, accountable, and confidential Cyber Threat Intelligence exchange within a permissioned, decentralized, non-blockchain environment. In this section, we discuss how DACTIE is situated in related work.

**Permissioned blockchain approaches.** TRADE [54] matches some requirements of DACTIE by keeping intelligence-sharing peers accountable while also ensuring anonymity. However, their concept limits anonymity to the communication between the CTI publisher and receiver. Additionally, the servers sharing the data possess excessive power; they can obtain information about the parties through their IP addresses and restrict or alter access to CTI, thereby compromising integrity. TRADE was enhanced by Dunnet et al. [49], utilizing AttriChain [55]. The framework allows anonymous sender identity and threshold traceability in a permissioned blockchain, which is even closer to DACTIE. However, neither solution has been implemented yet, as Allouche et al. have not advanced beyond the preprint stage, while Dunnett et al. are planning a PoC. Another permissioned blockchain network for CTI sharing is the one by Huff and Li [52]. They at least evaluated their Merkle tree maintenance and token authentication, but also provided no open-source PoC we could evaluate against.

**Public blockchain approaches.** The research group of Dunnett et al. has proposed several ideas combining CTI sharing with the public Ethereum blockchain [48], [64], [65]. All smart contract transactions incur a gas fee, which can be a few dollars per CTI shared or requested. While the authors argue that this cost is minor compared to potential losses from security breaches, it may not be practical. If sharing and requesting CTI incurs costs, user incentives may decrease. Furthermore, financial concerns are a primary reason companies avoid CTI platforms (Challenges C8, C10). As 24% of the participants of a study conducted by Fischer et al. state *"Lack of financing"* is the reason for not wanting to use threat intelligence sharing platforms now or in the future [18]. Additionally, costs increase with the volume of CTI consumed. For example, if each CTI request costs $4.45, organizations may hesitate to purchase critical information if their daily spending limit is already reached, potentially leaving them vulnerable to attacks.

**Non-blockchain approaches.** Open-source, CTI-sharing platforms, such as MISP, were not designed for anonymity and do not provide sending messages anonymously, although an open feature request for it exists [66]. The open-source platform OTX[13] by Alienvault does not provide sender anonymity, as interactions with the central platform can be tracked and linked to accounts - one of the major drawbacks of centralized architectures with a single point of control, which makes them inferior at promoting anonymity.

## 9. Conclusion

This paper presented DACTIE, a novel design and technical implementation of a decentralized, permissioned network specifically tailored for efficient, anonymous, accountable, and confidential cyber threat intelligence sharing. Through workshops with real-world stakeholders, including government, industry, and critical infrastructure organizations, we identified these requirements, which are essential for the acceptance of our approach. With our utilized technologies, including the GossipSub protocol, MLS, and our novel, adapted implementation of the Short Threshold Dynamic Group Signature scheme, we have developed a system that not only protects sender identities and metadata anonymity of receivers but also enhances trust through accountability, allowing for controlled de-anonymization of messages when necessary. The evaluation shows that DACTIE meets our requirements and performs better than related work built on HF. We demonstrate that our approach is an effective, decentralized CTI sharing solution advancing the state of the art.

13. https://otx.alienvault.com/

# Appendix

## Member Join Protocol and Example Message

Figure 7 shows a sequence diagram of the steps needed for a Member Candidate to join the DACTIE network.

Figure 8 illustrates two exemplary messages with identical content, sent by the same peer shortly one after another.

## Artifacts

This appendix lists the artifacts necessary to evaluate the contributions of this paper and describes how the review committees can access them. All artifacts are provided in anonymized form to preserve double-blind review.

### Artifact 1: Source Code and Experiments

- **Contents:** Implementation of the system and attacks described in the paper, including:
  - all peer types (authority, archive, peer)
  - ps16 and cam20 schemes
  - evaluation experiments: latency, eclipse, and anonymity
- **Access:** Public anonymized repository at:
  - https://anonymous.4open.science/r/Dactie-Anon/README.md
- **Usage:** A detailed `README.md` in the repository documents prerequisites and step-by-step commands to reproduce all figures and tables. In brief:
  - Requirements: Linux, Rust (newest release), Docker, and standard build tools.
  - Build: `cargo build` in n terminals for n peers.

### Artifact 2: Processed Experimental Results

- **Contents:**
  - processed CSV files from which we generated the tables and figures
- **Access:** Available in the same anonymized repository under:
  - `results/`
- **Usage:** To regenerate the figures from the stored results:

Listing 1. Latency Experiment Peer1

```
RUST_LOG=info cargo run --package
    dactie-peer --bin dactie-peer -- \
  -K /<folder-peer1>/ \
  -b <AUTH_ID> \
  -B /ip4/<ip-addr-authority>/tcp/10001
      \
  -p <peer1-port>  \
  --advertise-addr /ip4/<ip-addr-peer1
      >/tcp/<peer1-port> | tee /tmp/
      peer_bench.log

register <AUTH_ID> peer1

sub Test
```

Listing 2. Latency Experiment Peer2

```
BENCH_PUBLISH=true BENCH_RATE=10
    BENCH_TOPIC=Test \
RUST_LOG=info cargo run --package
    dactie-peer --bin dactie-peer -- \
  -K /<folder-peer2>/ \
  -b <AUTH_ID> \
  -B /ip4/<ip-addr-authority>/tcp/10001
      \
  -p <peer2-port> \
  --advertise-addr /<ip-addr-peer2>/tcp
      /<peer2-port>

register <AUTH_ID> peer1

sub Test
```

Listing 3. Eclipse Experiment

```
for A in 10 20 30 40 50 60 70 80 90
    100; do
    raw_n=$(( (A + 98) / 99 ))
    n=$(( raw_n < 2 ? 2 : raw_n ))
    N=$(( A / 4 ))
    cargo run \
        --package dactie-experiment-
            eclipse \
        --bin dactie-experiment-eclipse
            -- \
        -n "$n" -N "$N" -A "$A" -a 60 -
            s 180 \
        -r 3 -o ./output/${A}
done >> ./output.txt
```

Listing 4. Anonymity Experiment

```
for i in 100 200 300 400 500 600 700
    800 900 1000; do
    k=$((i - 100))
    echo ${k}
    cargo run \
        --package dactie-experiments \
        --bin dactie-experiments -- \
        -r 4 -t 240 -i fix=70 -n ${k} \
        -N ${i} -S 25 -o ./output/${i}
done >> ./output.txt
```

These commands regenerate all data used to create the respective CSV files, which is manually done afterward.

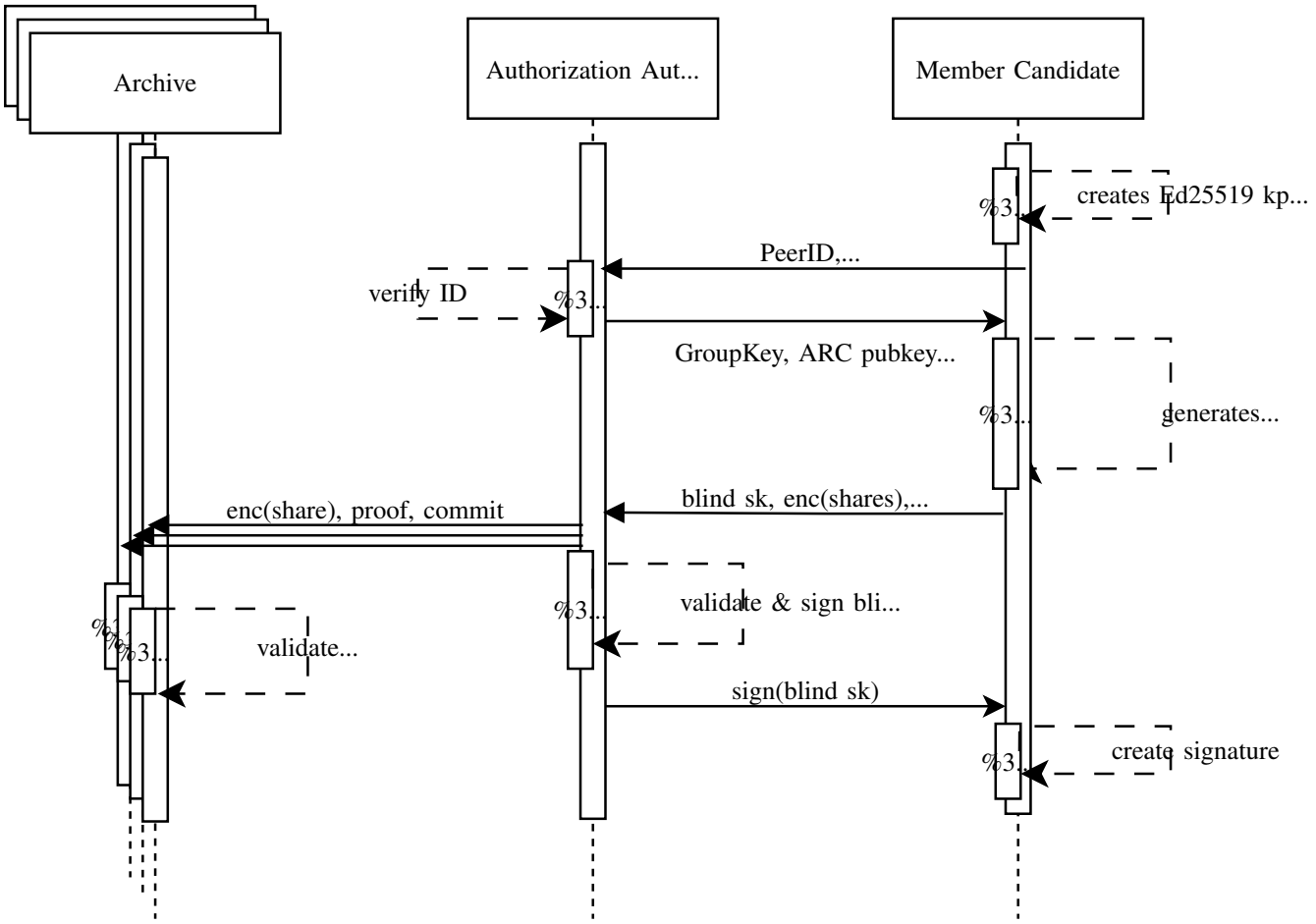



Figure 7. Member join protocol: sequence of steps and message exchanges required for a new peer to join the DACTIE network.

```
Got Broadcastmessage:
'"Some CTI regarding a firewall vulnerability"'
Topic: 'firewall'
Propagating Peer: 12D3KoolVMK2ZgtshopjwcyAJgNnokrP
Signature: c3000000000000002831373939393U33336U537U
38U1U3U13935UU32U33114331383643U538313038333138313
310142313231313...
Got Broadcastmessage:
'"Some CTI regarding a firewall vulnerability"'
Topic: 'firewall'
Propagating Peer: 12D3KoolVMK2ZgtshopjwcyAJgNnokrP
Signature: c3000000000000002830U13742U1U532303U383
363736333330U13232383033384U3132U43832U53540U33938
303536323830393...
```

Figure 8. Console output of two messages with the same PeerID, the same content, but different signatures received by another peer subscribed to the topic.

## References


[1] P. Alaeifar, S. Pal, Z. Jadidi, M. Hussain, and E. Foo, "Current approaches and future directions for Cyber Threat Intelligence sharing: A survey," *Journal of Information Security and Applications*, vol. 83, p. 103786, Jun. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214212624000899

[2] V. Mavroeidis and S. Bromander, "Cyber Threat Intelligence Model: An Evaluation of Taxonomies, Sharing Standards, and Ontologies within Cyber Threat Intelligence," in *2017 European Intelligence and Security Informatics Conference (EISIC)*. Athens: IEEE, Sep. 2017, pp. 91–98. [Online]. Available: http://ieeexplore.ieee.org/document/8240774/

[3] V. A. Countryman, "Cybersecurity Risk Management, Strategy, Governance, and Incident Disclosure," p. 186, Jul. 2023. [Online]. Available: https://www.sec.gov/newsroom/press-releases/2023-139

[4] 117th Congress, "H.R.2471 - Consolidated Appropriations Act, 2022," Mar. 2022. [Online]. Available: https://www.congress.gov/bill/117th-congress/house-bill/2471/text

[5] CISA, "Cyber Incident Reporting for Critical Infrastructure Act of 2022 (CIRCIA)," 2022. [Online]. Available: https://www.cisa.gov/topics/cyber-threats-and-advisories/information-sharing/cyber-incident-reporting-critical-infrastructure-act-2022-circia

[6] E. P. a. Council, "Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive) (Text with EEA relevance)," Dec. 2022. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32022L2555&qid=1675336594166

[7] E. Commision, "NIS2 Directive: new rules on cybersecurity of network and information systems," Jan. 2025. [Online]. Available: https://digital-strategy.ec.europa.eu/en/policies/nis2-directive

[8] P. of Australia, "Security Legislation Amendment (Critical Infrastructure Protection) Act 2022," Apr. 2022. [Online]. Available: https://www.legislation.gov.au/C2022A00033/asmade/text

[9] ISO/IEC, "ISO/IEC 27001:2022," 2022. [Online]. Available: https://www.iso.org/standard/27001

[10] R. Brown and A. Sfakianakis, "SANS CTI Survey 2024: Managing the Evolving Threat Landscape," SANS, Tech. Rep., May 2024. [Online]. Available: https://www.sans.org/white-papers/2024-cti-survey-managing-evolving-threat-landscape/

[11] ——, "SANS 2025 CTI Survey: Navigating Uncertainty in Today's Threat Landscape," SANS, Tech. Rep., May 2025. [Online]. Available: http://bit.ly/3UktUkd

[12] ENISA, "EU incident response and cyber crisis management." [Online]. Available: https://www.enisa.europa.eu/topics/eu-incident-response-and-cyber-crisis-management

[13] E. secretariat team, "CSIRTs network FAQ." [Online]. Available: https://csirtsnetwork.eu/homepage?tab=faq

[14] M. He, L. Devine, and J. Zhuang, "Perspectives on cybersecurity information sharing among multiple stakeholders using a decision-theoretic approach," *Risk Analysis*, vol. 38, no. 2, pp. 215–225, 2018. [Online]. Available: https://doi.org/10.1111/risa.12878

[15] C. Abraham, F. Bélanger, and S. Daultrey, "Promoting research on cyber threat intelligence sharing in ecosystems," *Journal of Cybersecurity*, vol. 11, no. 1, p. tyaf016, Jan. 2025. [Online]. Available: https://doi.org/10.1093/cybsec/tyaf016

[16] T. D. Wagner, K. Mahbub, E. Palomar, and A. E. Abdallah, "Cyber threat intelligence sharing: Survey and research directions," *Computers & Security*, vol. 87, p. 101589, Nov. 2019. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S016740481830467X

[17] C. S. Johnson, M. L. Badger, D. A. Waltermire, J. Snyder, and C. Skorupka, "Guide to Cyber Threat Information Sharing," National Institute of Standards and Technology, Tech. Rep. NIST SP 800-150, Oct. 2016. [Online]. Available: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-150.pdf

[18] D. Fischer, C. Sauerwein, M. Werchan, and D. Stelzer, "An Exploratory Study on the Use of Threat Intelligence Sharing Platforms in Germany, Austria and Switzerland," in *Proceedings of the 18th International Conference on Availability, Reliability and Security*. Benevento Italy: ACM, Aug. 2023, pp. 1–7. [Online]. Available: https://dl.acm.org/doi/10.1145/3600160.3600185

[19] S. Murdoch and N. Leaver, "Anonymity vs. Trust in Cyber-Security Collaboration," in *Proceedings of the 2nd ACM Workshop on Information Sharing and Collaborative Security*. Denver Colorado USA: ACM, Oct. 2015, pp. 27–29. [Online]. Available: https://dl.acm.org/doi/10.1145/2808128.2808134

[20] T. Schaberreiter, V. Kupfersberger, K. Rantos, A. Spyros, A. Papanikolaou, C. Ilioudis, and G. Quirchmayr, "A Quantitative Evaluation of Trust in the Quality of Cyber Threat Intelligence Sources," in *Proceedings of the 14th International Conference on Availability, Reliability and Security*. Canterbury CA United Kingdom: ACM, Aug. 2019, pp. 1–10. [Online]. Available: https://dl.acm.org/doi/10.1145/3339252.3342112

[21] D. Homan, I. Shiel, and C. Thorpe, "A New Network Model for Cyber Threat Intelligence Sharing using Blockchain Technology," in *2019 10th IFIP International Conference on New Technologies, Mobility and Security (NTMS)*. CANARY ISLANDS, Spain: IEEE, Jun. 2019, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/document/8763853/

[22] A. Giehl, P. Schneider, M. Busch, F. Schnoes, R. Kleinwort, and M. F. Zaeh, "Edge-computing enhanced privacy protection for industrial ecosystems in the context of smes," in *2019 12th CMI Conference on Cybersecurity and Privacy (CMI)*. IEEE, 2019, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8962138/

[23] A. Giehl, M. P. Heinl, and V. Embacher, "Emuflex: A flexible ot testbed for security experiments with opc ua," *Proceedings of the 19th International Conference on Availability, Reliability and Security*, pp. 1–9, 2024. [Online]. Available: https://dl.acm.org/doi/abs/10.1145/3664476.3670931

[24] A. Zibak and A. Simpson, "Cyber Threat Information Sharing: Perceived Benefits and Barriers," in *Proceedings of the 14th International Conference on Availability, Reliability and Security*. Canterbury CA United Kingdom: ACM, Aug. 2019, pp. 1–9. [Online]. Available: https://dl.acm.org/doi/10.1145/3339252.3340528

[25] T. Geras and T. Schreck, "Sharing Communities: The Good, the Bad, and the Ugly," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*. Copenhagen Denmark: ACM, Nov. 2023, pp. 2755–2769. [Online]. Available: https://dl.acm.org/doi/10.1145/3576915.3623144

[26] P. Baran, "On Distributed Communications: I. Introduction to Distributed Communications Networks," RAND Corporation., Tech. Rep. RM-3420-PR, Aug. 1964. [Online]. Available: https://www.rand.org/pubs/research_memoranda/RM3420.html

[27] D. Chaum and E. van Heyst, "Group Signatures," in *Advances in Cryptology — EUROCRYPT '91*, D. W. Davies, Ed. Berlin, Heidelberg: Springer, 1991, pp. 257–265. [Online]. Available: https://link.springer.com/content/pdf/10.1007/3-540-46416-6_22

[28] D. Pointcheval and O. Sanders, "Short Randomizable Signatures," in *Topics in Cryptology - CT-RSA 2016*, K. Sako, Ed. Springer International Publishing, 2016, vol. 9610, pp. 111–126. [Online]. Available: http://link.springer.com/10.1007/978-3-319-29485-8_7

[29] M. Nardelli, F. D. Sclavis, and M. Iezzi, "A Hitchhiker's Guide to Privacy-Preserving Digital Payment Systems: A Survey on Anonymity, Confidentiality, and Auditability," Oct. 2025, arXiv:2505.21008 [cs]. [Online]. Available: http://arxiv.org/abs/2505.21008

[30] J. Camenisch, M. Drijvers, A. Lehmann, G. Neven, and P. Towa, "Short threshold dynamic group signatures," in *International conference on security and cryptography for networks*. Springer, 2020, pp. 401–423. [Online]. Available: https://doi.org/10.1007/978-3-030-57990-6_20

[31] A. Shamir, "How to share a secret," *Commun. ACM*, vol. 22, no. 11, p. 612–613, Nov. 1979. [Online]. Available: https://doi.org/10.1145/359168.359176

[32] T. Perrin and M. Marlinspike, "The Double Ratchet Algorithm," Nov. 2016. [Online]. Available: https://signal.org/docs/specifications/doubleratchet/

[33] N. Unger, S. Dechand, J. Bonneau, S. Fahl, H. Perl, I. Goldberg, and M. Smith, "Sok: secure messaging," in *2015 IEEE Symposium on Security and Privacy*. IEEE, 2015, pp. 232–249. [Online]. Available: https://doi.org/10.1109/SP.2015.22

[34] T. Frosch, C. Mainka, C. Bader, F. Bergsma, J. Schwenk, and T. Holz, "How secure is textsecure?" *IEEE European Symposium on Security and Privacy (EuroS and P)*, p. 457–472, 2016. [Online]. Available: https://doi.org/10.1109/EuroSP.2016.41

[35] R. Barnes, B. Beurdouche, R. Robert, J. Millican, E. Omara, and K. Cohn-Gordon, "Rfc 9420: The messaging layer security (mls) protocol," 2023.

[36] D. Soler, C. Dafonte, M. Fernández-Veiga, A. F. Vilas, and F. J. Nóvoa, "Experimental Analysis of Efficiency of the Messaging Layer Security for Multiple Delivery Services," Feb. 2025, arXiv:2502.18303 [cs]. [Online]. Available: http://arxiv.org/abs/2502.18303

[37] FIRST, "TRAFFIC LIGHT PROTOCOL (TLP) FIRST Standards Definitions and Usage Guidance," Aug. 2022. [Online]. Available: https://www.first.org/tlp/

[38] D. L. Chaum, "Untraceable electronic mail, return addresses, and digital pseudonyms," *Association for Computing Machinery (ACM)*, vol. 24, no. 2, 1981. [Online]. Available: https://dl.acm.org/doi/pdf/10.1145/358549.358563

[39] A. M. Piotrowska, J. Hayes, T. Elahi, S. Meiser, and G. Danezis, "The Loopix Anonymity System," in *Proceedings of the 26th USENIX Security Symposium: August 16-18, 2017, Vancouver, BC, Canada*. Berkeley, CA: USENIX Association, Aug. 2017.

[40] C. Kuhn, M. Beck, S. Schiffner, E. Jorswieck, and T. Strufe, "On privacy notions in anonymous communication," 2019. [Online]. Available: https://arxiv.org/abs/1812.05638

[41] D. Chaum, "Security without identification: transaction systems to make big brother obsolete," *Communications of the ACM*, vol. 28, no. 10, pp. 1030–1044, Oct. 1985. [Online]. Available: https://dl.acm.org/doi/10.1145/4372.4373

[42] R. Barnes, B. Beurdouche, R. Robert, J. Millican, E. Omara, and K. Cohn-Gordon, "The Messaging Layer Security (MLS) Protocol," RFC 9420, Jul. 2023. [Online]. Available: https://www.rfc-editor.org/info/rfc9420

[43] D. Vyzovitis and Y. Psaras, "Gossipsub: A secure pubsub protocol for unstructured, decentralised p2p overlays," *Proceedings of the Protocol Labs TechRep, PL-TechRep-gossipsub-v0*, 2019. [Online]. Available: https://research.protocol.ai/blog/2019/a-new-lab-for-resilient-networks-research/PL-TechRep-gossipsub-v0.1-Dec30.pdf

[44] R. Guerraoui, A.-M. Kermarrec, A. Kucherenko, R. Pinot, and S. Voitovych, "On the inherent anonymity of gossiping," *arXiv preprint arXiv:2308.02477*, 2023. [Online]. Available: https://arxiv.org/pdf/2308.02477

[45] Eng Keong Lua, J. Crowcroft, M. Pias, R. Sharma, and S. Lim, "A survey and comparison of peer-to-peer overlay network schemes," *IEEE Communications Surveys & Tutorials*, vol. 7, no. 2, pp. 72–93, 2005. [Online]. Available: https://ieeexplore.ieee.org/document/1610546/

[46] S. Henningsen, S. Rust, M. Florian, and B. Scheuermann, "Crawling the ipfs network," in *IFIP Networking 2020 Conference and Workshops*. Paris, France: IEEE, Jun. 2020.

[47] I. Baumgart and S. Mies, "S/Kademlia: A practicable approach towards secure key-based routing," in *2007 International Conference on Parallel and Distributed Systems*. Hsinchu, Taiwan: IEEE, 2007, pp. 1–8. [Online]. Available: http://ieeexplore.ieee.org/document/4447808/

[48] K. Dunnett, S. Pal, G. D. Putra, Z. Jadidi, and R. Jurdak, "A trusted, verifiable and differential cyber threat intelligence sharing framework using blockchain," in *2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2022, pp. 1107–1114. [Online]. Available: https://doi.org/10.1109/TrustCom56396.2022.00152

[49] K. Dunnett, S. Pal, Z. Jadidi, G. D. Putra, and R. Jurdak, "A democratically anonymous and trusted architecture for cti sharing using blockchain," in *2022 International Conference on Computer Communications and Networks (ICCCN)*, 2022, pp. 1–7. [Online]. Available: https://doi.org/10.1109/ICCCN54977.2022.9868919

[50] W. Bugden and A. Alahmar, "Rust: The programming language for safety and performance," 2022. [Online]. Available: https://arxiv.org/abs/2206.05503

[51] J. Camenisch, M. Drijvers, A. Lehmann, G. Neven, and P. Towa, "Short Threshold Dynamic Group Signatures," in *Security and Cryptography for Networks*, C. Galdi and V. Kolesnikov, Eds. Cham: Springer International Publishing, 2020, vol. 12238, pp. 401–423, series Title: Lecture Notes in Computer Science. [Online]. Available: https://link.springer.com/10.1007/978-3-030-57990-6_20

[52] P. Huff and Q. Li, "A Distributed Ledger for Non-attributable Cyber Threat Intelligence Exchange," in *Security and Privacy in Communication Networks*, J. Garcia-Alfaro, S. Li, R. Poovendran, H. Debar, and M. Yung, Eds. Cham: Springer International Publishing, 2021, vol. 398, pp. 164–184, series Title: Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering. [Online]. Available: https://link.springer.com/10.1007/978-3-030-90019-9_9

[53] P. Thakkar, S. Nathan, and B. Viswanathan, "Performance Benchmarking and Optimizing Hyperledger Fabric Blockchain Platform," in *2018 IEEE 26th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS)*. Milwaukee, WI: IEEE, Sep. 2018, pp. 264–276. [Online]. Available: https://ieeexplore.ieee.org/document/8526892/

[54] Y. Allouche, N. Tapas, F. Longo, A. Shabtai, and Y. Wolfsthal, "Trade: trusted anonymous data exchange: threat sharing using blockchain technology," Mar. 2021, arXiv:2103.13158 [cs]. [Online]. Available: http://arxiv.org/abs/2103.13158

[55] W. Shao, C. Jia, Y. Xu, K. Qiu, Y. Gao, and Y. He, "Attrichain: Decentralized traceable anonymous identities in privacy-preserving permissioned blockchain," *Computers and Security*, vol. 99, p. 102069, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167404820303424

[56] Y. Psaras, "Gossipsub Message Propagation Latency - Networking - Ethereum Research." [Online]. Available: https://ethresear.ch/t/gossipsub-message-propagation-latency/19982

[57] S. Lenz, "Evaluation of the messaging layer security protocol: A performance and usability study," 2020. [Online]. Available: https://www.diva-portal.org/smash/get/diva2:1388449/FULLTEXT01.pdf

[58] R. Frank, O. Lomeli, N. Giridharan, S. Ponnapalli, M. K. Aguilera, and N. Crooks, "Real life is uncertain. consensus should be too!" in *Proceedings of the 2025 Workshop on Hot Topics in Operating Systems*, ser. HotOS '25. New York, NY, USA: Association for Computing Machinery, 2025, p. 80–87. [Online]. Available: https://doi.org/10.1145/3713082.3730374

[59] R. Canetti, E. Kushilevitz, R. Ostrovsky, and A. Rosén, "Randomness versus fault-tolerance," vol. 13, no. 1, pp. 107–142. [Online]. Available: http://link.springer.com/10.1007/s001459910005

[60] M. Ben-Or, S. Goldwasser, and A. Wigderson, "Completeness theorems for non-cryptographic fault-tolerant distributed computation," in *Proceedings of the Twentieth Annual ACM Symposium on Theory of Computing*, ser. STOC '88. New York, NY, USA: Association for Computing Machinery, 1988, p. 1–10. [Online]. Available: https://doi.org/10.1145/62212.62213

[61] J. Alwen, S. Coretti, Y. Dodis, and Y. Tselekounis, "Modular design of secure group messaging protocols and the security of mls," in *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*, 2021, pp. 1463–1483.

[62] J. Alwen, D. Jost, and M. Mularczyk, "On the insider security of mls," in *Annual International Cryptology Conference*. Springer, 2022, pp. 34–68. [Online]. Available: https://doi.org/10.1145/3460120.3484820

[63] D. Vyzovitis, Y. Napora, D. McCormick, D. Dias, and Y. Psaras, "GossipSub: Attack-Resilient Message Propagation in the Filecoin and ETH2.0 Networks." [Online]. Available: http://arxiv.org/abs/2007.02754

[64] K. Dunnett, S. Pal, Z. Jadidi, and R. Jurdak, "A blockchain-based framework for scalable and trustless delegation of cyber threat intelligence," in *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2023, pp. 1–9. [Online]. Available: https://doi.org/10.1109/ICBC56567.2023.10174885

[65] K. Dunnett, S. Pal, Z. Jadidi, V. Dedeoglu, and R. Jurdak, "Privshare: A privacy-preserving framework for differential and trustless delegation of cyber threat intelligence using blockchain," *Computer Networks*, vol. 252, p. 110686, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1389128624005188

[66] mxm0z, "Feature Request: Anonymization of Org Names within Sharing Groups · Issue #10019 · MISP/MISP," Oct. 2024. [Online]. Available: https://github.com/MISP/MISP/issues/10019